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Inventor(s)

LIEN; Bei-Shing et al.

INTEGRATED CIRCUIT AND METHOD OF OPERATING THE SAME

Abstract

An integrated circuit is provided, which includes a first temperature-sensitive device and a second temperature-sensitive device. The first temperature-sensitive device is configured to generate a first bias current which monotonically increases in accordance with an absolute temperature, and generate a first voltage based on the first bias current. The second temperature-sensitive device is configured to generate a second voltage across the second temperature-sensitive device, and output a reference voltage at an output terminal of the integrated circuit. The second voltage monotonically decreases in accordance with the absolute temperature. The reference voltage equals the first voltage plus the second voltage.

Inventors: LIEN; Bei-Shing (Hsinchu, TW), LIU; Szu-Lin (Hsinchu, TW)

Applicant: TAIWAN SEMICONDUCTOR MANUFACTURING COMPANY, LTD.
(Hsinchu, TW)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application claims the benefit of U.S. Provisional Application No. 63/562,466, filed Mar. 7, 2024, the entire disclosure of which is incorporated by reference herein.

BACKGROUND

[0002] The current trend in miniaturizing integrated circuits (ICs) has led to the development of smaller, more efficient devices with increased functionality and higher operating speeds. This miniaturization process has also brought about more stringent design and manufacturing requirements, as well as reliability challenges. Electronic design automation (EDA) tools are utilized to create, optimize, and validate standard cell layout designs for integrated circuits, ensuring that they meet both design and manufacturing specifications.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is emphasized that, in accordance with standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features can be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1A is a schematic diagram of a voltage reference circuit in accordance with some embodiments of the present disclosure.

[0005] FIG. 1B is a diagram of a stacked gate device in accordance with some embodiments of the present disclosure.

[0006] FIG. 1C is an equivalent circuit diagram of the stacked gate device in FIG. 1B in accordance with some embodiments of the present disclosure.

[0007] FIG. 1D is a diagram of a stacked gate device with multiple finger structures in accordance with some embodiments of the present disclosure.

[0008] FIG. 1E is an equivalent circuit diagram of the stacked gate device in FIG. 1D in accordance with some embodiments of the present disclosure.

[0009] FIG. 2A is a schematic diagram of a stacked gate device in a diode-connected configuration in accordance with some embodiments of the present disclosure.

[0010] FIG. 2B is a diagram illustrating variations of a voltage-temperature curve of the stacked gate device in FIG. 2A in accordance with some embodiments of the present disclosure.

[0011] FIG. 3A is a schematic diagram of a cascode structure in accordance with some embodiments of the present disclosure.

[0012] FIG. 3B is a diagram illustrating a voltage-temperature curve of the cascode structure in FIG. 3A in accordance with some embodiments of the present disclosure.

[0013] FIG. 4 is a schematic diagram of a voltage reference circuit in accordance with some embodiments of the present disclosure.

[0014] FIGS. 5A-5D are schematic diagrams of trimming stacked gate devices with different number of finger structures in accordance with some embodiments of the present disclosure.

[0015] FIGS. 6A-6D are schematic diagrams of a buffer circuit in different implementations in accordance with some embodiments of the present disclosure.

[0016] FIG. 7 is a flowchart of a method for operating a voltage reference circuit in accordance

with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0017] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features can be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0018] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “over,” “upper,” “on” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0019] Further, it will be understood that when an element is referred to as being “connected to” or “coupled to” another element, it can be directly connected to or coupled to the other element, or intervening elements can be present.

[0020] Embodiments, or examples, illustrated in the drawings are disclosed as follows using specific language. It will nevertheless be understood that the embodiments and examples are not intended to be limiting. Any alterations or modifications in the disclosed embodiments, and any further applications of the principles disclosed in this document are contemplated as would normally occur to one of ordinary skill in the pertinent art.

[0021] Further, it is understood that several processing steps and/or features of a device can be only briefly described. Also, additional processing steps and/or features can be added, and certain of the following processing steps and/or features can be removed or changed while still implementing the claims. Thus, it is understood that the following descriptions represent examples only, and are not intended to suggest that one or more steps or features are required.

[0022] In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0023] In some embodiments, a voltage reference circuit is implemented to generate a reference voltage using stacked gate devices. A stacked gate device includes a plurality of field-effect transistors having a common gate terminal, and having channels connected in series. A first temperature-sensitive device is implemented based on stacked gate devices to generate a first bias current which monotonically increases with an absolute temperature of the voltage reference circuit, and to generate a first voltage based on the first bias current. A second temperature-sensitive device is implemented based on the stacked gate devices to generate a second voltage, which monotonically decreases with the absolute temperature of the voltage reference circuit, across the second temperature-sensitive device. The temperature dependency of the reference voltage generated by the voltage reference circuit can be compensated using the first voltage generated by the first temperature-sensitive device and the second voltage generated by the second temperature-sensitive device.

[0024] FIG. 1A is a schematic diagram of a voltage reference circuit in accordance with some embodiments of the present disclosure. FIG. 1B is a diagram of a stacked gate device in accordance

with some embodiments of the present disclosure. FIG. 1C is an equivalent circuit diagram of the stacked gate device in FIG. 1B in accordance with some embodiments of the present disclosure. [0025] In some embodiments, the voltage reference circuit **100** is a bandgap voltage reference circuit which is configured to provide a reference voltage VREF. In some embodiments, the reference voltage VREF is equal to supply voltage VDD. In some embodiments, the reference voltage VREF is equal to a voltage between the supply voltage VDD and the reference voltage VSS. The voltage reference circuit **100** may include transistors **M1**, **M2** and **M3**, temperature-sensitive devices **110** and **120**, and a trimming circuit **130**, as depicted in FIG. 1A. The transistors **M1**, **M2** and **M3** may be field-effect transistors (FETs). Each of the transistors **M1**, **M2** and **M3** has a gate terminal and a channel between a source/drain (S/D) terminal and a S/D terminal. The current passing through the channel depends on the voltage difference applied to the gate terminal of each of transistor **M1**, **M2** and **M3**.

[0026] The voltage reference circuit **100** includes stacked gate devices **X1**, **X2**, **X3**, **X1B**, **X2B**, and **X3_trim0** to **X3_trimx** where x is an integer corresponding to the number of stacked gate trim devices. Each of the stacked gate devices **X1**, **X2**, **X3**, **X1B**, and **X2B** includes a plurality of field-effect transistors (i.e., referred as “transistors” hereafter) stacked together. The references **X1**, **X2**, **X3**, **X1B** and **X2B** are also used to represent the number of FETs in each of the respective stacked gate devices **X1**, **X2**, **X3**, **X1B** and **X2B**. In some embodiments, each of the stacked gate devices **X3_trim0** to **X3_trimx** have the same number of stacked transistors as the stacked gate device **X3**, but the number of finger structures of the stacked gate devices **X3_trim0** to **X3_trimx** may differ from that of the stacked gate device **X3**. The details of a stacked gate device are described as follows.

[0027] In some embodiments, a stacked gate device **150**, also known as “stack X” in FIG. 1B, is a three-terminal transistor device with a gate terminal **151**, a S/D terminal **152**, and a S/D terminal **153**. The equivalent circuit diagram of the stacked gate device **150** includes a plurality of transistors **1501** arranged in a cascode structure or a stacked structure, as shown in FIG. 1C. The total number of stacked transistors **1501** is denoted as an integer X. For example, the gate terminals of the transistors **1501** are connected together to form the gate terminal **151** of the stacked gate device **150**. Additionally, the transistors **1501** may be N-type FETs, and the N-type channels of the transistors **1501** (e.g., X transistors **1501**) are connected in series between the S/D terminal **152** and the S/D terminal **153** of the stacked gate device **150**. For example, the S/D terminal of the first transistor **1501** serves as the S/D terminal **152** of the stacked gate device **150**, and a S/D terminal of the first transistor **1501** is connected to a S/D terminal of the second transistor **1501**, a S/D terminal of the second transistor **1501** is connected to a S/D terminal of the third transistor **1501**, . . . , and so on. In other words, for each integer n between 1 to X-1, the S/D terminal of the n-th transistor **1501** is connected to the S/D terminal of the (n+1)-th transistor **1501**. Accordingly, the S/D terminal of the last transistor **1501** (i.e., X-th transistor **1501**) serves as the S/D terminal **153** of the stacked gate device **150**.

[0028] FIG. 1D is a diagram of a stacked gate device with multiple finger structures in accordance with some embodiments of the present disclosure. FIG. 1E is an equivalent circuit diagram of the stacked gate device in FIG. 1D in accordance with some embodiments of the present disclosure.

[0029] In some embodiments, the stacked gate device **150** of FIG. 1B may include one or more stacked gate devices **TX1** to **TXN** arranged in parallel, as shown in FIG. 1D, where N is a positive integer corresponding to the number of parallel stacked gate devices. In some embodiments, each of the stacked gate devices **TX1** to **TXN** is referred to as a finger structure or a “finger”, which includes X transistors **1501** arranged in a cascode structure or a stacked structure, as shown in FIG. 1E. For example, the channel of the transistors **1501** within each stacked gate device **TX1** to **TXN** are connected in series to form the respective channel of each stacked gate device **TX1** to **TXN**. Additionally, the channel of each stacked gate device **TX1** to **TXN** is coupled between the S/D terminal **152** and S/D terminal **153** of the stacked gate device **150**, while the gate terminals of

stacked gate devices TX1 to TXN are connected to the gate terminal **151** of the stacked gate device **150**. When the stacked gate device **150** includes one finger structure, the equivalent circuit diagram of the stacked gate device **150** can be referred to FIG. **1C**.

[0030] In some embodiments, the transistors **1501** within the stacked gate devices TX1 to TXN may be fabricated within the same process, and thus have substantially the same electrical characteristics, such as channel width, channel length, threshold voltage, and transconductance. In some embodiments, the design of a stacked gate device X with one or more finger structures shown in FIGs. **1D** and **1E** can be applied in a similar manner to the stacked gate devices X1, X2, X3, X1B, X2B, and X3_trim0 to X3_trimx in FIG. **1A**, and similar detailed description is omitted.

[0031] In some embodiments, the temperature-sensitive device **110** includes a cascode structure **111** and a cascode structure **112**, as depicted in FIG. **1A**. The current Ib1 generated by the temperature-sensitive device **110** monotonically increases in accordance with an absolute temperature (e.g., proportional to the absolute temperature (PTAT)) of the voltage reference circuit **100**, resulting in the voltage VPTAT generated by the temperature-sensitive device **110** at node N3 to monotonically increase in accordance with the absolute temperature of the voltage reference circuit **100**. Accordingly, the temperature-sensitive device **110** can be referred to as a PTAT device. The details thereof will be described with reference to FIGs. **3A** to **3B**.

[0032] In some embodiments, the cascode structure **111** includes stacked gate devices X1 and X2, while the cascode structure **112** includes stacked gate devices X1B and X2B. Each of the gate terminals of the stacked gate devices X1 and X2, the gate terminals of the stacked gate devices X1B and X2B the S/D terminal of the stacked gate device X2 are electrically connected together at node N2. The S/D terminal of the stacked gate device X2 is electrically connected to the S/D terminal of the stacked gate device X1. The S/D terminal of the stacked gate device X1 is connected to the ground voltage (or common reference voltage) VSS.

[0033] In some embodiments, the temperature-sensitive device **120** includes the stacked gate device X3. Each of the gate terminal and S/D terminal of the stacked gate device X3, the S/D terminal of transistor M3, the S/D terminal of the plurality of trimming stacked gate devices X3_trim0 to X3_trimx are electrically connected together at node N4 which is an output terminal of the voltage reference circuit **100** configured to provide the reference voltage VREF. Each of the S/D terminal of the stacked gate device X3, the S/D terminal of the stacked gate device X1 and the S/D terminal of the stacked gate device X2 and the S/D terminal of the plurality of trimming stacked gate devices X3_trim0 to X3_trimx are electrically connected together at node N3. The voltage (e.g., Vgs of X3) generated by the temperature-sensitive device **120** monotonically decreases in accordance with the absolute temperature (e.g., complementary to the absolute temperature (CTAT)) of the voltage reference circuit **100**. In some embodiments, the temperature-sensitive device **120** can be referred to as a CTAT device. The details thereof will be described with reference to FIGs. **2A** to **2B**.

[0034] In some embodiments, the trimming circuit **130** is configured to adjust (e.g., fine-tune) the matching between the voltage-temperature rising rate of the temperature-sensitive device **110** and the voltage-temperature falling rate of the temperature-sensitive device **120** by a dynamic element matching (“DEM”) technique. The trimming circuit **130** may include a plurality of trimming stacked gate devices X3_trim0 to X3_trimx. The gate terminal of each trimming stacked gate device X3_trim0 to X3_trimx is coupled to a respective bit of a trimming code signal TC[0] . . . TC[x] through a corresponding buffer circuit FB0 to FBx. The S/D terminal and S/D terminal of each trimming stacked gate device X3_trim0 to X3_trimx is coupled between node N4 and node N3, which provide the reference voltage VREF and the voltage VPTAT, respectively. Additionally, each of the buffer circuits FB0 to FBx may be supplied with the reference voltage VREF and the ground voltage VSS, as shown in FIG. **1A**.

[0035] In some embodiments, each of the trimming stacked gate devices X3_trim0 to X3_trimx includes one or more finger structures arranged in parallel, where each finger structure has an equal

number of stacked transistors as the stacked gate device **X3**. In some embodiments, the trimming stacked gate devices **X3_trim0** to **X3_trimx** can have an equal number or different numbers of finger structures as one another, depending on the type of the trimming code signal **TC[0] . . . TC[x]**. The details thereof will be further described with reference to FIGS. 5A to 5D.

[0036] In some embodiments, the gate terminals of transistor **M1**, **M2**, and **M3** are electrically connected to node **N1**, and the S/D terminals of transistors **M1**, **M2**, and **M3** are electrically connected to a power supply voltage **VDD**. Since transistors **M1** and **M2** have the same gate-to-source voltage **V_{gs}**, transistors **M1** and **M2** may be configured to function as a first current mirror (not labelled), and the bias current **I_{b2}** passing through the channel of transistor **M2** is proportional to the bias current **I_{b1}** passing through the channel of transistor **M1**. In some embodiments, when transistors **M1** and **M2** are designed with substantially the same electrical characteristics, such as channel width, channel length, threshold voltage, and transconductance, the bias current **I_{b2}** flowing through transistor **M2** is substantially equal to the bias current **I_{b1}** flowing through transistor **M1**. Thus, in these embodiments, transistor **M1** may function as a current source, and the bias current **I_{b1}** flows through transistor **M1**, the cascode structure **112**, and a resistor **R**. In some embodiments, the resistor **R** is the resistance of the voltage reference circuit **100**. Similarly, transistor **M2** may function as another current source, and the bias current **I_{b2}** flows through transistor **M2** and the cascode structure **111**.

[0037] In some embodiments, since transistors **M1** and **M3** have the same gate-to-source voltage **V_{gs}**, transistors **M1** and **M3** may be configured to function as a second current mirror (not labelled), and the bias current **I_{b3}** passing through the channel of transistor **M3** is proportional to the bias current **I_{b1}** passing through the channel of transistor **M1**. In some embodiments, when transistors **M1** and **M3** are designed with substantially the same electrical characteristics, such as channel width, channel length, threshold voltage, and transconductance, the bias current **I_{b3}** flowing through transistor **M3** is substantially equal to the bias current **I_{b1}** flowing through transistor **M1**. For example, transistor **M3** may function as yet another current source, and the bias current **I_{b3}** flows through transistor **M3** of the temperature-sensitive device **120**, and the stacked gate device **X1**. Therefore, a total current of **I_{b2}+I_{b3}**, which is substantially twice the bias current **I_{b1}**, flows through the stacked gate device **X1**, as depicted in FIG. 1A.

[0038] FIG. 2A is a schematic diagram of a stacked gate device in a diode-connected configuration in accordance with some embodiments of the present disclosure. FIG. 2B is a diagram illustrating variations of a voltage-temperature curve of the stacked gate device in FIG. 2A in accordance with some embodiments of the present disclosure.

[0039] In some embodiments, the stacked gate device **150** is in a diode-connected configuration, indicating that the gate terminal **151** of a stacked gate device **150** is connected to the S/D terminal **152** of the stacked gate device **150**, and a bias current **I_b** is supplied to the stacked gate device **150**, as shown in FIG. 2A. In such a case, the voltage difference **V_{gs}** between the gate terminal **151** and S/D terminal **153** of the stacked gate device **150** decreases as the absolute temperature of the stacked gate device **X** increases, as shown by curve **202** in FIG. 2B. Additionally, the downward slope of the voltage-temperature curve depends on the number of stacked transistors within the stacked gate device **X**. For example, as the number of stacked transistors increases (e.g., a larger stack **X**), the slope of the V-T curve decreases, as shown by curve **204** in FIG. 2B, indicating that the voltage difference **V_{gs}** between the gate terminal **151** and the S/D terminal **153** of the stacked gate device **X** becomes less sensitive to changes in absolute temperature. As a result, the output voltage **VOI** generated by the stacked gate device **150** monotonically decreases in accordance with the absolute temperature (e.g., complementary to the absolute temperature (CTAT)). Accordingly, the stacked gate device **150** in the configuration shown in FIG. 2A is configured to operate as a CTAT device, in accordance with some embodiments. Furthermore, since the configuration of the stacked gate device **X3** shown in FIG. 1A is similar to that of the stacked gate device **150** shown in FIG. 2A, the stacked gate device **X3** can be regarded as a CTAT device, in accordance with some

embodiments.

[0040] FIG. 3A is a schematic diagram of a cascode structure in accordance with some embodiments of the present disclosure. FIG. 3B is a diagram illustrating a voltage-temperature curve of the cascode structure in FIG. 3A in accordance with some embodiments of the present disclosure.

[0041] In some embodiments, the cascode structure **111** shown in FIG. 1A is a PTAT device which can be implemented by the stacked gate devices **X1** and **X2**. The gate terminals of the stacked gate devices **X1** and **X2** are connected to node **N2**. The stacked gate devices **X1** and **X2** may be single threshold voltage (e.g., single V_t) stacked gate devices, which indicates that the transistors within the stacked gate devices **X1** and **X2** have substantially the same threshold voltage V_t , resulting in high PTAT correlation between the gate-to-source voltage V_{gs_X2} of the stacked gate device **X2** and the gate-to-source voltage V_{gs_X1} of the stacked gate device **X1**. As described in the embodiments of FIGS. 2A-2B, both the gate-to-source voltage V_{gs_X2} and the gate-to-source voltage V_{gs_X1} decreases as the temperature increases.

[0042] In some embodiments, the downward slope of the V-T curve of the stacked gate device **X2** can become less steep as the number of stacked transistors within the stacked gate device **X2** increases. Similarly, in some embodiments, the downward slope of the V-T curve of the stacked gate device **X1** can also become less steep as the number of stacked transistors within the stacked gate device **X1** increases. In some embodiments, when the number of stacked transistors within the stacked gate device **X1** is larger than that within the stacked gate device **X2**, the cascode structure **111** can function as a PTAT device. For example, the output voltage V_{PTAT} of the cascode structure **111** can be calculated as $V_{gs_X1} - V_{gs_X2}$. As the temperature increases, the decrement of the gate-to-source voltage V_{gs_X2} of the stacked gate device **X2** is greater than the decrement of the gate-to-source voltage V_{gs_X1} of the stacked gate device **X1**, such that the voltage difference $V_{gs_X1} - V_{gs_X2}$ increases. Accordingly, the voltage V_{PTAT} at node **N3** versus the temperature can be plotted as a V-T curve **310**, as shown in FIG. 3B, which has an upward slope. In some embodiments,, the upward slope of curve **310** depends on the difference $X1 - X2$ between the numbers **X1** and **X2**. In some embodiments, the upward slope of curve **310** increases as the difference $X1 - X2$ increases. In some embodiments, the output voltage V_{PTAT} of the cascode structure **111**, which is proportional to the absolute temperature, can be used to compensate the output voltage (e.g., V_{gs_X3}), which is complementary to the absolute temperature, generated by the temperature-sensitive device **120** (e.g., stacked gate device **X3**), allowing the voltage reference circuit **100** to generate the reference voltage V_{REF} with a temperature coefficient (TC) close to zero.

[0043] Attention now is directed back to FIG. 1A. In some embodiments, the cascode structure **111** works together with the cascode structure **112** to enable the temperature-sensitive device **110** to function as a PTAT device. The cascode structure **112** is similar to the cascode structure **111**, and similar detailed description is omitted. For example, the number **X1B** of stacked transistors within the stacked gate device **X1B** is equal to the number **X1** of stacked transistors within the stacked gate device **X1** (i.e., $X1 = X1B$), while the number **X2B** of stacked transistors within the stacked gate device **X2B** is equal to the number **X2** of stacked transistors within the stacked gate device **X2** (i.e., $X2 = X2B$). In some embodiments, since the overall bias current $I_{b2} + I_{b3}$ flowing through the stacked gate device **X1** is approximately twice the bias current I_{b1} flowing through the stacked gate devices **X1B** and **X2B** of the cascode structure **112**, the stacked gate device **X1** can be designed to have twice the number of finger structures as the stacked gate device **X1B**. In some embodiments, the stack gate devices **X2** and **X2B** have the same number of finger structures as each other.

[0044] In some embodiments, the overall size (e.g., width of channels) of finger structures within the stacked gate device **X1B** and **X2B** is larger than that within the stacked gate device **X1** and **X2**. Accordingly, the gate-to-source voltage V_{gs_X1BX2B} of the stacked gate devices **X1B** and **X2B**

within the cascode structure **112** is lower than the gate-to-source voltage V_{gs_X1X2} of the stacked gate devices **X1** and **X2** within the cascode structure **111**. Furthermore, the bias current I_{b1} flowing through the resistor **R** can be expressed by equation (1) as follows.

$$[00001] \quad I_{b1} = \frac{V_{gs_X1X2} - V_{gs_X1BX2B}}{R} \quad (1)$$

[0045] In some embodiments, since the overall size of finger structures within the stacked gate device **X1B** and **X2B** is larger than that within the stacked gate device **X1** and **X2**, the decrement rate of the gate-to-source voltage V_{gs_X1X2} is lower than that of the gate-to-source voltage V_{gs_X1BX2B} as the temperature increases. Accordingly, the voltage difference ($V_{gs_X1X2} - V_{gs_X1BX2B}$) is proportional to the absolute temperature, and thus the bias current I_{b1} flowing through the resistor **R** is a PTAT current. Furthermore, since transistors **M1** and **M2** forms a current mirror, the bias current I_{b2} generated by transistor **M2** is also a PTAT current. Similarly, since transistor **M1** and **M3** forms another current mirror, the bias current I_{b3} generated by transistor **M3** is also a PTAT current.

[0046] More specifically, the bias current I_{b3} , which is a PTAT current, flows through the temperature-sensitive device **120** (e.g., stacked gate device **X3**). As the temperature increases, the bias current I_{b3} increases, leading to an increase of the gate-to-source voltage V_{gs_X3} . That is, the gate-to-source voltage V_{gs_X3} is PTAT. However, as described in the embodiments of FIGS. 2A-2B, since the stacked gate device **X3** is in a diode-connected configuration, the stacked gate device **X3** can be regarded as a CTAT device. On the other hand, the voltage V_{PTAT} generated by the temperature-sensitive device **110** at node **N3** is PTAT, as described in the embodiments of FIGS. 3A-3B. Accordingly, for the temperature-sensitive device **120**, the PTAT scheme (e.g., I_{b3} and V_{PTAT}) can be compensated with the CTAT scheme (e.g., V_{gs_X3}), allowing the voltage reference circuit **100** to generate the reference voltage V_{REF} with a temperature coefficient substantially equal to 0 at node **N4**, in accordance with some embodiments.

[0047] FIG. 4 is a schematic diagram of a voltage reference circuit in accordance with some embodiments of the present disclosure.

[0048] In some embodiments, the CTAT scheme can be adjusted by the trimming circuit **130**. For brevity, the trimming circuit **130** within the voltage reference circuit **100B** shown in FIG. 4 may include four trimming stacked gate devices **X3_trim0** to **X3_trim3** that are controlled by respective bits of the trimming code signal $TC[0:3]$ through respective buffer circuits **FB0** to **FB3**. In some embodiments, the buffer circuits **FB0** to **FB3** are supplied with the reference voltage V_{REF} and the ground voltage **VSS**. Additionally, each bit of the trimming code signal $TC[0:3]$ may be passed to the gate terminals **B0** to **B3** of the trimming stacked gate devices **X3_trim0** to **X3_trim3** through the respective buffer circuits **FB0** to **FB3**. In some embodiments, the voltage range of each bit of the trimming code signal $TC[0:3]$ is between the reference voltage V_{REF} and the ground voltage **VSS**.

[0049] In some embodiments, each of the trimming stacked gate device **X3_trim0** to **X3_trim3** may have the same number of finger structures, such as 1 to **N**, where **N** is a positive integer. When thermal meter coding is used for the trimming circuit **130**, each bit of the trimming code signal $TC[0:3]$ can control an equal number of finger structures to couple to the stacked gate device **X3** in parallel. For brevity, it is assumed that each of the trimming stacked gate device **X3_trim0** to **X3_trim3** has one finger structure, in accordance with some embodiments. When the trimming code signal $TC[3:0]$ is equal to 1101, the reference voltage V_{REF} is passed to the gate terminals **B0**, **B2**, and **B3** of the trimming stacked gate device **X3_trim0**, **X3_trim2** and **X3_trim3**, activating the trimming stacked gate devices **X3_trim0**, **X3_trim2** and **X3_trim3**. Meanwhile, the ground voltage **VSS** is passed to the gate terminal **B1**, deactivating the trimming stacked gate device **X3_trim1**. Accordingly, a total number of 3 finger structures are activated and coupled in parallel with each other, and further coupled to the stacked gate device **X3** so as to adjust the downward slope of the V-T curve of the stacked gate device **X3**, thereby performing temperature-coefficient

trimming on the reference voltage VREF generated by the voltage reference circuit **100B**.

[0050] In some embodiments, each of the trimming stacked gate devices X3_trim0 to X3_trim3 may have different numbers of finger structures, such as powers of 2. For brevity, it is assumed that there are 4 stacked transistors within the stacked gate device X3. The trimming stacked gate devices X3_trim0 to X3_trim3 include 1, 2, 4, and 8 finger structures, respectively, with each finger structure including 4 stacked transistors, as shown in FIGS. 5A-5D, in accordance with some embodiments. When binary coding is used for the trimming circuit **130**, each bit of the trimming code signal TC[0:3] can control different numbers of finger structures to be coupled together in parallel and further coupled to the stacked gate device X3 in parallel. When the trimming code signal TC[3:0] is equal to 1101, the reference voltage VREF is passed to the gate terminals B0, B2, and B3 of the trimming stacked gate device X3_trim0, X3_trim2, and X3_trim3, activating the trimming stacked gate device X3_trim0, X3_trim2, and X3_trim3. Meanwhile, the ground voltage VSS is passed to the gate terminal B1, deactivating the trimming stacked gate device X3_trim1. Accordingly, a total number of 13 (e.g., 1+4+8) finger structures are activated to couple to the stacked gate device X3 in parallel so as to adjust the downward slope of the V-T curve of the stacked gate device X3, thereby performing temperature-coefficient trimming on the reference voltage VREF generated by the voltage reference circuit **100B**.

[0051] FIGS. 6A-6D are schematic diagrams of a buffer circuit in different implementations in accordance with some embodiments of the present disclosure.

[0052] In some embodiments, at least one or more of the buffer circuits FB0 to FBx in FIG. 1A can be implemented using the buffer circuit **600A** shown in FIG. 6A. The buffer circuit **600A** includes two inverters (e.g., transistors Q1+Q2 and Q3+Q4) connected in series, that are supplied with the reference voltage VREF and the ground voltage VSS. The input signal TC[x] of the buffer circuit **600A** can be passed to the gate terminal Bx of the trimming stacked gate device X3_trimx through the buffer circuit **600A**. Additionally, when the input signal TC[x] is in the high logic state and the low logic state, the input signal TC[x] can be at the reference voltage VREF and the ground voltage VSS, respectively. For example, in response to the input signal TC[x] being in the high logic state (e.g., "1"), transistor Q2 is turned on and transistor Q1 is turned off, causing the voltage at node N7 to be pulled down to the ground voltage VSS. At this time, transistor Q3 is turned on and transistor Q4 is turned off, causing the voltage at the gate terminal Bx to be pulled up to the reference voltage VREF. As a result, the trimming stacked gate device X3_trimx is turned on (e.g., selected), and the one or more finger structures within the trimming stacked gate device X3_trimx are coupled to the stacked gate device X3 in parallel, indicating that the selected trimming stacked gate device X3_trimx can contribute to the V-T curve of the temperature-sensitive device **120**.

[0053] On the other hand, in response to the input signal TC[x] being in the low logic state (e.g., "0"), transistor Q1 is turned on and transistor Q2 is turned off, causing the voltage at node N7 to be pulled up to the reference voltage VREF. At this time, transistor Q4 is turned on and transistor Q3 is turned off, causing the voltage at the gate terminal Bx to be pulled down to the ground voltage VSS. As a result, the trimming stacked gate device X3_trimx is turned off (e.g., unselected), and the one or more finger structures within the trimming stacked gate device X3_trimx are not coupled to the stacked gate device X3, indicating that the unselected trimming stacked gate device X3_trimx has no influence on the V-T curve of the temperature-sensitive device **120**.

[0054] In some embodiments, each of the buffer circuits FB0 to FBx in FIG. 1A can be implemented using the buffer circuit **600B** shown in FIG. 6B. The input signal of the buffer circuit **600B** may be TC[x]' which is complementary to the respective bit TC[x] of the trimming signal TC. For example, in response to the input signal TC[x]' being in the high logic state (e.g., "1"), transistor Q2 is turned on and transistor Q1 is turned off, causing the voltage at the gate terminal Bx to be pulled down to the ground voltage VSS. As a result, the trimming stacked gate device X3_trimx is turned off (e.g., unselected), and the one or more finger structures within the trimming stacked gate device X3_trimx are not coupled to the stacked gate device X3, indicating that the

unselected trimming stacked gate device **X3_trimx** has no influence on the V-T curve of the temperature-sensitive device **120**. On the other hand, in response to the input signal **TC[x]'** being in the low logic state (e.g., "0"), transistor **Q1** is turned on and transistor **Q2** is turned off, causing the voltage at the gate terminal **Bx** to be pulled up to the reference voltage **VREF**. As a result, the trimming stacked gate device **X3_trimx** is turned on (e.g., selected), and the one or more finger structures within the trimming stacked gate device **X3_trimx** are coupled to the stacked gate device **X3** in parallel, indicating that the selected trimming stacked gate device **X3_trimx** can contribute to the V-T curve of the temperature-sensitive device **120**.

[0055] In some embodiments, at least one or more of the buffer circuits **FB0** to **FBx** in FIG. **1A** can be implemented using the buffer circuit **600C** shown in FIG. **6C**. The buffer circuit **600C** may be implemented using a CMOS transmission gate which includes transistors **Q5** and **Q6**. In response to the input signals **TC[x]** and **TC[x]'** being respectively in the high logic state (e.g., "1") and the low logic state (e.g., "0"), transistors **Q5** and **Q6** are turned on, causing the reference voltage **VREF** to be passed to the gate terminal **Bx** through the buffer circuit **600C**. As a result, the trimming stacked gate device **X3_trimx** is turned on (e.g., selected), and the one or more finger structures within the trimming stacked gate device **X3_trimx** are coupled to the stacked gate device **X3** in parallel, indicating that the selected trimming stacked gate device **X3_trimx** can contribute to the V-T curve of the temperature-sensitive device **120**. On the other hand, in response to the input signals **TC[x]** and **TC[x]'** being respectively in the low logic state (e.g., "0") and the high logic state (e.g., "1"), transistors **Q5** and **Q6** are turned off, causing the gate terminal **Bx** to be in a floating state. When the gate terminal **Bx** of the trimming stacked gate device **X3_trimx** is floating, the trimming stacked gate device **X3_trimx** is turned off (e.g., unselected), and the one or more finger structures within the trimming stacked gate device **X3_trimx** are not coupled to the stacked gate device **X3**, indicating that the unselected trimming stacked gate device **X3_trimx** has no influence on the V-T curve of the temperature-sensitive device **120**.

[0056] In some embodiments, at least one or more of the buffer circuits **FB0** to **FBx** in FIG. **1A** can be implemented using the buffer circuit **600D** shown in FIG. **6D**. The buffer circuit **600D** includes switches **S1** and **S2** that are respectively controlled by the input signals **TC[x]** and **TC[x]'**. In response to the input signals **TC[x]** and **TC[x]'** being respectively in the high logic state (e.g., "1") and the low logic state (e.g., "0"), switch **S1** is activated and switch **S2** is deactivated, causing the reference voltage **VREF** to be passed to the gate terminal **Bx** through switch **S1**. As a result, the trimming stacked gate device **X3_trimx** is turned on (e.g., selected), and the one or more finger structures within the trimming stacked gate device **X3_trimx** are coupled to the stacked gate device **X3** in parallel, indicating that the selected trimming stacked gate device **X3_trimx** can contribute to the V-T curve of the temperature-sensitive device **120**. On the other hand, in response to the input signals **TC[x]** and **TC[x]'** being respectively in the low logic state (e.g., "0") and the high logic state (e.g., "1"), switch **S1** is deactivated and switch **S2** is activated, causing the voltage at the gate terminal **Bx** to be pulled down to the ground voltage **VSS**. As a result, the trimming stacked gate device **X3_trimx** is turned off (e.g., unselected), and the one or more finger structures within the trimming stacked gate device **X3_trimx** are not coupled to the stacked gate device **X3**, indicating that the unselected trimming stacked gate device **X3_trimx** has no influence on the V-T curve of the temperature-sensitive device **120**.

[0057] FIG. **7** is a flowchart of a method for operating a voltage reference circuit in accordance with some embodiments of the present disclosure. The sequence in which the operations of method **700** are depicted in FIG. **7** is for illustration only; the operations of method **700** are capable of being executed in sequences that differ from that depicted in FIG. **7**. It is understood that additional operations may be performed before, during, and/or after the method **700** depicted in FIG. **7**, and that some other processes may only be briefly described herein.

[0058] In operation **710**, a first bias current is generated to pass through a first cascode structure within a first temperature-sensitive device and a resistor. In the embodiment of FIG. **1A**, the

temperature-sensitive device **110** generates the bias current Ib1 (e.g., the first bias current) passing through the cascode structure **112**, which includes stacked gate devices X1B and X2B, and the resistor R. Additionally, the bias current Ib1 (e.g., the first bias current) is proportional to the absolute temperature.

[0059] In operation **720**, a second bias current is generated to pass through a second cascode structure within the first temperature-sensitive device based on the first bias current. In the embodiment of FIG. **1A**, transistors M1 and M2 form a first current mirror, and the bias current Ib2 (e.g., the second bias current) is generated by transistor M2 based on the bias current Ib1. Since the bias current Ib1 is PTAT, the bias current Ib2 is also PTAT. Additionally, the cascode structure **111** (e.g., second cascode structure) includes the stacked gate device X1 and X2.

[0060] In operation **730**, a third bias current is generated to pass through a second temperature-sensitive device and a first stacked gate device within the second cascode structure based on the first bias current. In the embodiment of FIG. **1A**, the second temperature-sensitive device includes the stacked gate device X3, and the bias current Ib3 (e.g., the third bias current) passes through the stacked gate device X3 and the first stacked gate device X1. Additionally, the stacked gate device X3 is a CTAT device.

[0061] In operation **740**, a reference voltage generated at a terminal of the second temperature-sensitive device is outputted. In the embodiment of FIG. **1A**, the reference voltage VREF is generated at the S/D terminal (e.g., node N4) of the stacked gate device X3. The reference voltage VREF is the sum of the gate-to-source of the stacked gate device X3 and the voltage VPTAT generated by the temperature-sensitive device **110** at node N3.

[0062] In operation **750**, turning on a set of trimming stacked gate devices in response to input signals TC. In some embodiments, operation **750** includes feeding back the reference voltage to the set of trimming stacked devices, and turning on the set of trimming stacked gate devices in response to input signals TC. In some embodiments, the set of trimming stacked gate devices includes one or more of trimming stacked gate devices X3_trim0 to X3_trimx. In some embodiments, one or more selected trimming stacked gate device(s) in the set of trimming stacked gate devices can contribute to the V-T curve of the temperature-sensitive device **120** thereby adjusting the V-T curve of the temperature-sensitive device **120** consistent with the description of the present disclosure described herein. In some embodiments, operation **750** includes performing temperature-coefficient trimming on the reference voltage VREF by the set of trimming stacked devices (e.g., the trimming circuit **130**). In some embodiments, operation **750** includes adjusting the CTAT scheme by the set of trimming stacked devices (e.g., the trimming circuit **130**).

[0063] An aspect of the present disclosure provides an integrated circuit, which includes a first temperature-sensitive device and a second temperature-sensitive device. The first temperature-sensitive device is configured to generate a first bias current which monotonically increases in accordance with an absolute temperature, and generate a first voltage based on the first bias current. The second temperature-sensitive device is configured to generate a second voltage across the second temperature-sensitive device, and output a reference voltage at an output terminal of the integrated circuit. The second voltage monotonically decreases in accordance with the absolute temperature. The reference voltage equals the first voltage plus the second voltage.

[0064] Another aspect of the present disclosure provides an integrated circuit, which includes a first cascode structure, a second cascode structure, a first current mirror, a second current mirror, and a first stacked gate device. The first cascode structure is configured to generate a first bias current which is proportional to an absolute temperature. The first current mirror is configured to generate a second bias current in response to the first bias current. The second cascode structure is coupled to the first cascode structure, and configured to generate a first voltage at a first node in response to the second bias current. The second current mirror is configured to generate a third bias current in response to the first bias current. The third bias current is proportional to the absolute temperature. The first stacked gate device is coupled between the first node and an output terminal of the

integrated circuit, and configured to receive the third bias current, generate a second voltage across the first stacked gate device, and output a reference voltage at the output terminal of the integrated circuit. The second voltage is complementary to the absolute temperature. The reference voltage equals the first voltage plus the second voltage.

[0065] Yet another aspect of the present disclosure provides a method. The method includes the following steps: generating a first bias current passing through a first cascode structure and a resistor; generating, based on the first bias current, a second bias current passing through a second cascode structure; generating, based on the first bias current, a third bias current passing through a temperature-sensitive device and a first stacked gate device within the second cascode structure; and outputting a reference voltage generated at a terminal of the temperature-sensitive device.

[0066] The methods and features of the present disclosure have been sufficiently described in the provided examples and descriptions. It should be understood that any modifications or changes without departing from the spirit of the present disclosure are intended to be covered in the protection scope of the present disclosure.

[0067] Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, and composition of matter, means, methods and steps described in the specification. As those skilled in the art will readily appreciate from the present disclosure, processes, machines, manufacture, composition of matter, means, methods or steps presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein, can be utilized according to the present disclosure.

[0068] Accordingly, the appended claims are intended to include within their scope processes, machines, manufacture, compositions of matter, means, methods or steps. In addition, each claim constitutes a separate embodiment, and the combination of various claims and embodiments are within the scope of the present disclosure.

Claims

1. An integrated circuit, comprising: a first temperature-sensitive device configured to generate a first bias current which monotonically increases in accordance with an absolute temperature of the integrated circuit, and generate a first voltage based on the first bias current; and a second temperature-sensitive device configured to generate a second voltage across the second temperature-sensitive device, and output a reference voltage at an output terminal of the integrated circuit, wherein the second voltage monotonically decreases in accordance with the absolute temperature of the integrated circuit, wherein the reference voltage is equal to a sum of the first voltage and the second voltage.
2. The integrated circuit of claim 1, wherein the first voltage is proportional to the absolute temperature of the integrated circuit, and the second voltage is complementary to the absolute temperature of the integrated circuit.
3. The integrated circuit of claim 1, wherein the first temperature-sensitive device comprises: a first cascode structure comprising: a first stacked gate device having a gate terminal connected to a first node, a first terminal connected to a second node, and a second terminal connected to a ground voltage; and a second stacked gate device having a gate terminal connected to the first node, a first terminal connected to the first node, and a second terminal connected to the second node; and a second cascode structure comprising: a third stacked gate device having a gate terminal connected to the first node, a first terminal connected to a third node, and a second terminal connected to a fourth node; and a fourth stacked gate device having a gate terminal connected to the first node, a first terminal connected to a fifth node, and a second terminal connected to the third node.
4. The integrated circuit of claim 3, further comprising: a resistor coupled between the fourth node and the ground voltage.

5. The integrated circuit of claim 4, wherein: the first stacked gate device comprises a plurality of first finger structures arranged in parallel with each other, wherein each first finger structure of the plurality of first finger structures comprises a first number of field-effect transistors connected in series; the second stacked gate device comprises one or more second finger structures arranged in parallel with each other, wherein each second finger structure of the one or more second finger structures comprises a second number of field-effect transistors connected in series; and the first number is greater than the second number.

6. The integrated circuit of claim 5, wherein: the third stacked gate device comprises one or more third finger structures arranged in parallel with each other, wherein each third finger structure of the one or more third finger structures comprises the first number of field-effect transistors connected in series; the fourth stacked gate device comprises one or more fourth finger structures arranged in parallel with each other, wherein each fourth finger structure of the one or more fourth finger structures comprises the second number of field-effect transistors connected in series; a number of the first finger structures is twice that of the third finger structures; and a number of the second finger structures is equal to the fourth finger structures.

7. The integrated circuit of claim 6, wherein an overall width of the third finger structures and the fourth finger structures is greater than an overall width of the first finger structures and the second finger structures.

8. The integrated circuit of claim 6, wherein the field-effect transistors within the first stacked gate device, the second stacked gate device, the third stacked gate device and the fourth stacked gate device have a substantially equal threshold voltage to each other.

9. The integrated circuit of claim 6, wherein: the second temperature-sensitive device further comprises a fifth stacked gate device having a gate terminal connected to the output terminal of the integrated circuit, a first terminal connected to the output terminal of the integrated circuit, and a second terminal connected to the second node; and the fifth stacked gate device comprises one or more fifth finger structures, with each fifth finger structure comprising a third number of field-effect transistors connected in series.

10. The integrated circuit of claim 9, further comprising: a trimming circuit comprising: a plurality of trimming stacked gate devices arranged in parallel to the fifth stacked gate device; and a plurality of buffer circuits; wherein each of the trimming stacked gate devices is configured to receive a corresponding bit of a trimming code signal by a corresponding one of the buffer circuits.

11. The integrated circuit of claim 10, wherein each of the trimming stacked gate devices of the plurality of trimming stacked gate devices comprises a different number of finger structures in powers of 2, and each finger structure within each of the trimming stacked gate devices of the plurality of trimming stacked gate devices comprises the third number of field-effect transistors connected in series.

12. The integrated circuit of claim 10, wherein each of the trimming stacked gate devices of the plurality of trimming stacked gate devices comprises an equal number of finger structures, and each finger structure within each of the trimming stacked gate devices of the plurality of trimming stacked gate devices comprises the third number of field-effect transistors connected in series.

13. The integrated circuit of claim 10, wherein in response to a first bit of a first trimming stacked gate device of the plurality of trimming stacked gate devices being in a first logic state, the reference voltage is supplied to a gate terminal of the first trimming stacked gate device of the plurality of trimming stacked gate devices through a first buffer circuit of the plurality of buffer circuits thereby enabling the first trimming stacked gate device of the plurality of trimming stacked gate devices to be electrically coupled to the fifth stacked gate device in parallel.

14. The integrated circuit of claim 13, wherein in response to the first bit of the first trimming stacked gate device of the plurality of trimming stacked gate devices being in a second logic state complementary to the first logic state, the ground voltage is supplied to the gate terminal of the first trimming stacked gate device of the plurality of trimming stacked gate devices through the first

buffer circuit of the plurality of buffer circuits thereby disabling the first trimming stacked gate device of the plurality of trimming stacked gate devices from being electrically coupled to the fifth stacked gate device in parallel.

15. An integrated circuit, comprising: a first cascode structure configured to generate a first bias current that is proportional to an absolute temperature of the integrated circuit; a first current mirror configured to generate a second bias current in response to the first bias current; a second cascode structure coupled to the first cascode structure, and configured to generate a first voltage at a first node in response to the second bias current; a second current mirror configured to generate a third bias current in response to the first bias current, wherein the third bias current is proportional to the absolute temperature of the integrated circuit; and a first stacked gate device coupled between the first node and an output terminal of the integrated circuit, and configured to receive the third bias current, generate a second voltage across the first stacked gate device, and output a reference voltage at the output terminal of the integrated circuit; wherein the second voltage is complementary to the absolute temperature of the integrated circuit, wherein the reference voltage is equal to a sum of the first voltage and the second voltage.

16. The integrated circuit of claim 15, further comprising: a resistor, wherein the first bias current is configured to flow from a power supply voltage to a ground voltage through the first current mirror, the second cascode structure, and the resistor.

17. The integrated circuit of claim 16, wherein a third voltage across the first cascode structure is greater than a fourth voltage across the second cascode structure.

18. The integrated circuit of claim 17, wherein: the first cascode structure comprises: a second stacked gate device having a gate terminal connected to a second node, a first terminal connected to the first node, and a second terminal connected to the ground voltage; and a third stacked gate device having a gate terminal connected to the second node, a first terminal connected to the second node, and a second terminal connected to the first node; and the second cascode structure comprises: a fourth stacked gate device having a gate terminal connected to the second node, a first terminal connected to a third node, and a second terminal connected to the resistor; and a fifth stacked gate device, having a gate terminal connected to the second node, a first terminal connected to the first current mirror, and a second terminal connected to the third node.

19. A method, comprising: generating a first bias current passing through a first cascode structure and a resistor; generating, based on the first bias current, a second bias current passing through a second cascode structure; generating, based on the first bias current, a third bias current passing through a temperature-sensitive device and a first stacked gate device within the second cascode structure; and outputting a reference voltage generated at a terminal of the temperature-sensitive device.

20. The method of claim 19, further comprising: generating a first voltage by the second cascode structure in response to the second bias current; and generating, by the temperature-sensitive device, a second voltage across the temperature-sensitive device; wherein the first voltage is proportional to an absolute temperature, and the second voltage is complementary to the absolute temperature, wherein the reference voltage is a sum of the first voltage and the second voltage.
